Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Non-linear representations in quantum mechanics

Files.

- File: `briefs/NonLinearQuantum/NonLinearQuantum.lean`
- Library: `NonLinearQuantum`
- Imports: `Mathlib.Analysis.InnerProductSpace.Calculus`, `Mathlib.Analysis.SpecialFunctions.ExpDeriv`

Setting. $H = \mathbb{C}^2$ (`EuclideanSpace ℂ (Fin 2)`), with $|z^+\rangle = e_0$, $|z^-\rangle = e_1$, and

$$m(v) = e^{i\theta\, p(z^+|v)}\, v, \qquad p(z^+ \mid v) = \frac{|v_0|^2}{\|v\|^2}.$$

In Lean $m(0) = 0$ holds without a case split, since $0/0 = 0$.

Results.

Brief	Lean
eq. (3), rays to rays	`nlMap_smul`: $m(kv) = k\,m(v)$ for every $k \in \mathbb{C}$
each ray is mapped to itself	`nlMap_mem_ray`
eq. (4), m not linear	`nlMap_not_add`, `not_isLinearMap_nlMap`, for $\theta \in (0, 2\pi)$
eq. (5), inner product	`inner_nlMap`
eq. (6), Born rule preserved	`born_nlMap`
Remark: m bijective	`nlHomeomorph`: a homeomorphism of $\mathbb{C}^2$, inverse $m_{-\theta}$
Remark: m a diffeomorphism (correction)	`contDiffAt_nlMap`: smooth at every $v \neq 0$
	`not_differentiableAt_nlMap_zero`: not differentiable at 0
Theorem 1, mathematical content	`exists_nonlinear_born_preserving`

Differences from the brief. Theorem 1 ("REP-VSLIN is independent of QST") is a statement about the logical independence of two conditions. The checked content is the existence of a homeomorphism of H that maps every ray to itself, preserves the Born rule and is not additive.

Problems found in the brief. The closing remark says that m is a diffeomorphism. It is not differentiable at 0: along every real line $m(tv) = t\,m(v)$, so a derivative at 0 would have to equal m itself, and m is not additive. The correct statement is that m is a homeomorphism of $\mathbb{C}^2$ and a diffeomorphism of $\mathbb{C}^2 \setminus \{0\}$. Theorem 1 only needs a bijection, so it is not affected.